ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2

Unit 2.01: Sequences, Series & Sigma Notation

Name: _____

Q1. Consider an arithmetic sequence where the first term is a_1 = 5 and the common difference is d = 3 . What is the value of the 4th term, a_4 ?

[ ] A: 14
[ ] B: 17
[ ] C: 8
[ ] D: 11

Q2. A geometric sequence has a_1=6 and common ratio r=2 . Find a_{5} .

[ ] A: 96
[ ] B: 192
[ ] C: 98
[ ] D: -96

Q3. Consider a geometric sequence with positive common ratio r > 0 where the first term is a_1 = frac{1}{2} . If the sum of the first three terms is S_3 = frac{7}{2} , find the value of r .

[ ] A: 2
[ ] B: -2
[ ] C: 1/2
[ ] D: 3

Q4. A geometric sequence has a_1=2 and common ratio r=3 . Find a_{5} .

[ ] A: 162
[ ] B: 486
[ ] C: 165
[ ] D: -162

Q5. A geometric sequence has a_1=6 and common ratio r=3 . Find a_{5} .

[ ] A: 486
[ ] B: 1458
[ ] C: 489
[ ] D: -486

Q6. A geometric sequence has a_1=2 and common ratio r=-2 . Find a_{4} .

[ ] A: -16
[ ] B: 32
[ ] C: -18
[ ] D: 16

Q7. A geometric sequence has a_1=5 and common ratio r=3 . Find a_{3} .

[ ] A: 45
[ ] B: 135
[ ] C: 48
[ ] D: -45

Q8. A geometric sequence has a_1=1 and common ratio r=-3 . Find a_{5} .

[ ] A: 81
[ ] B: -243
[ ] C: 78
[ ] D: -81

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